

EDINBURGH GOGARBANK, United Kingdom

WMO: 031660

Lat: 55.9283N

Lon: 3.3444W

Elev: 57

StdP: 100.64

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.4	-6.3	2.2	-1.9	-5.0	2.5	-0.7	14.7	7.6	13.3	8.4	1.4	230	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	22.5	17.0	20.9	16.0	19.6	15.3	17.9	21.2	16.9	19.7	16.1	18.5	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.8	19.4	15.7	11.2	18.4	14.9	10.6	17.4	50.6	21.2	47.6	19.7	45.2	18.6	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.2	9.1	-6.6	26.1	2.5	2.4	-8.4	27.8	-9.8	29.1	-11.2	30.5	-13.0	32.2		
			-7.0	19.9	2.5	1.3	-8.7	20.9	-10.2	21.7	-11.6	22.4	-13.4	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	4.3	4.6	5.8	8.2	10.6	13.4	15.2	14.8	13.1	9.9	6.4	4.6
	DBStd	4.81	3.10	2.90	2.85	2.57	2.56	2.06	2.11	1.81	2.25	2.70	3.10	3.78
	HDD10.0	872	176	153	134	65	22	1	0	0	3	35	112	171
	HDD18.3	3314	434	386	389	304	239	147	101	111	159	261	357	427
	CDD10.0	603	1	1	3	11	42	104	161	148	94	32	4	2
	CDD18.3	5	0	0	0	0	0	1	3	1	1	0	0	0
	CDH23.3	34	0	0	0	0	2	4	26	2	1	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	4.1	4.8	4.3	4.7	4.3	4.1	3.6	3.4	3.6	4.0	3.9	4.2	4.6
Precipitation	PrecAvg	661	60	44	49	41	51	51	57	59	63	68	62	59
	PrecMax	848	120	112	91	93	136	133	156	120	150	133	197	129
	PrecMin	438	18	3	16	6	9	6	5	4	7	14	9	9
	PrecStd	101	27	27	23	20	29	30	34	31	38	34	37	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	13.2	15.4	18.6	22.0	23.4	26.2	22.8	21.7	17.6	14.8	13.4
		MCWB	10.8	10.2	10.8	12.8	16.1	17.5	18.7	17.4	16.6	14.5	12.8	11.5
	2%	DB	11.1	11.5	12.9	15.7	18.9	20.9	23.2	20.9	19.6	16.2	13.4	12.3
		MCWB	9.7	9.7	9.9	11.1	13.8	16.0	17.6	16.2	15.4	13.7	11.7	10.7
	5%	DB	10.1	10.4	11.5	14.1	16.7	19.2	21.3	19.7	18.3	15.2	12.2	11.2
		MCWB	8.6	8.7	8.9	10.2	12.5	14.8	16.4	15.4	14.7	13.0	10.6	9.7
	10%	DB	9.1	9.2	10.2	12.9	15.2	17.8	19.6	18.7	17.1	14.1	11.1	10.1
		MCWB	7.7	7.5	8.0	9.4	11.4	13.8	15.5	14.8	13.9	12.0	9.5	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.2	11.9	13.4	17.0	18.4	19.7	18.5	17.7	15.2	13.2	11.7
		MCDB	12.1	12.2	14.4	17.0	21.2	21.9	25.3	21.8	20.8	16.7	14.4	13.0
	2%	WB	9.8	9.8	10.4	11.9	14.5	16.5	18.2	17.2	16.3	14.2	11.9	10.8
		MCDB	11.0	11.2	12.4	15.2	17.6	19.8	22.1	19.6	18.6	15.8	13.2	12.1
	5%	WB	8.7	8.8	9.2	10.7	13.2	15.3	17.1	16.4	15.3	13.2	10.8	9.8
		MCDB	9.9	10.3	11.1	13.5	15.8	18.3	20.3	18.6	17.5	14.9	12.0	11.1
	10%	WB	7.7	7.6	8.2	9.7	12.0	14.4	16.2	15.6	14.4	12.3	9.6	8.6
		MCDB	9.1	9.1	10.1	12.3	14.5	17.1	18.8	17.7	16.4	14.0	10.9	10.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.4	6.2	7.3	7.8	7.6	8.0	7.6	7.1	6.4	5.6	5.2
		MCDBR	5.5	6.0	7.6	9.7	10.2	10.3	11.3	9.1	8.7	6.9	6.2	5.9
		MCWBR	5.1	4.9	5.2	5.9	6.0	6.0	6.1	5.2	5.3	5.2	5.5	5.6
	5% WB	MCDBR	5.5	5.8	7.0	8.7	9.2	9.1	10.0	8.0	7.5	6.4	6.2	5.9
		MCWBR	5.2	5.0	5.3	5.7	5.8	5.7	5.9	5.2	5.2	5.3	5.7	5.7
Clear-Sky Solar Irradiance	taub	0.324	0.338	0.369	0.389	0.389	0.388	0.388	0.373	0.360	0.343	0.318	0.307	
	taud	2.238	2.333	2.279	2.252	2.301	2.334	2.340	2.389	2.436	2.419	2.318	2.214	
	Ebn at Noon	564	723	787	827	851	856	847	837	797	712	576	483	
	Edh at Noon	57	75	100	118	119	117	114	102	85	68	52	46	
All-Sky Solar Radiation	RadAvg	0.52	1.16	2.19	3.57	4.71	4.60	4.43	3.56	2.53	1.38	0.67	0.38	
	RadStd	0.08	0.11	0.25	0.39	0.33	0.42	0.44	0.27	0.19	0.16	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page